Which Method Should I Use?

A practical decision guide for orientation data in archaeoastronomy and ethnoastronomy

Start with the **research question**, not the software menu.

Step 1 - What kind of data do I have?

Directions or azimuths

Use circular statistics.

Axes with no meaningful start/end direction

Treat them as axial data before applying directional methods.

Astronomical declinations

For the workshop curvigram, declination is treated as a linear angular coordinate over the analysed range.

Step 2 - What question am I asking?

Research question	First workshop method	What it tells you	Main caution
What does the pattern look like?	Individual-direction plot / rose diagram	Shape, modes, spread	Appearance depends partly on graphical choices
What is the typical direction?	Circular mean	Central direction	Can mislead for multimodal data
How concentrated are directions?	R-bar	Net directional concentration	Low R-bar can result from opposing clusters
How dispersed are directions?	Circular variance / circular SD	Circular spread	Interpret together with the plot
How precise is the mean direction?	Bootstrap CI	Precision of estimated mean	Not archaeological certainty
Is there a unimodal preferred direction rather than uniformity?	Rayleigh test	Evidence against uniformity for a unimodal alternative	Can miss bimodal/multimodal structure
Is one bell-like circular model useful?	von Mises model	Mean direction mu and concentration kappa	Do not force on obvious multimodality
How do uncertain declinations combine into a continuous curve?	Curvigram	Smoothed aggregate pattern	Kernel and bandwidth affect details
Is a curvigram feature robust?	Sensitivity analysis	Stability across reasonable settings	Not a formal significance test
Do two subgroups look different?	Separate plots and summaries	Descriptive differences	Formal between-group inference is outside this workshop

Step 3 - Recommended workflow

1. **Check the data**
    - units;
    - geographic convention;
    - directional versus axial;
    - missing values;
    - sample size;
    - measurement uncertainty.

2. **Plot before testing**
   - unimodal?
   - bimodal?
   - multimodal?
   - near-uniform?
   - isolated observations?
3. **Describe**
   - n;
   - mean direction if meaningful;
   - R-bar;
   - circular variance / SD;
   - confidence interval where appropriate.
4. **Test the question you actually have**
   - Rayleigh for a simple unimodal preferred-direction question;
   - do not treat it as a universal random/non-random button.
5. **Model only when useful**
   - a single von Mises distribution is appropriate only when one concentrated mode is scientifically plausible.
6. **Represent uncertainty explicitly**
   - in the curvigram exercise, each observation has its own reported plus/minus uncertainty.
7. **Test robustness**
   - change kernel and bandwidth deliberately;
   - report the specifications;
   - trust stable features more than specification-sensitive details.

Minimal R cookbook

```r
library(circular)

az <- circular(
  dat$azimuth_deg,
  units = "degrees",
  template = "geographics",
  modulo = "2pi"
)

rose.diag(az, bins = 18)

as.numeric(mean(az)) %% 360
rho.circular(az)
1 - rho.circular(az)

rayleigh.test(az)

mle.vonmises(az)
```

Three rules

Rule 1: Plot first, test second.

Rule 2: Match the method to the question and distribution shape.

Rule 3: Statistical regularity does not automatically establish cultural intention or an astronomical target.